

■ 3.2 Mathematical Functions

■ 3.2.1 Naming Conventions

Mathematical functions in *Mathematica* are given names according to definite rules. As with most *Mathematica* functions, the names are usually complete English words, fully spelled out. For a few very common functions, *Mathematica* uses the traditional abbreviations. Thus the modulo function, for example, is `Mod`, not `Modulo`.

Mathematical functions that are usually referred to by a person's name have names in *Mathematica* of the form *PersonSymbol*. Thus, for example, the Legendre polynomials $P_n(x)$ are denoted `LegendreP[n, x]`.

Although this convention does lead to longer function names, it avoids any ambiguity or confusion. If you use a particular function a lot, you can always define an alias for its name, as discussed in Section 2.7.1.

When the standard notation for a mathematical function involves both subscripts and superscripts, the subscripts are given *before* the superscripts in the *Mathematica* form. Thus, for example, the associated Legendre polynomials $P_n^m(x)$ are denoted `LegendreP[n, m, x]`.